SAT Math Mastery · Session 2

Nonlinear Functions

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics quadratic models: vertex, axis of symmetry, maximum and minimum · writing a quadratic from given conditions · transformations and composition · exponential growth and decay · equivalent exponential forms · comparing functions · factors and zeros · reading graphs and tables · rational and radical equations · inverse functions · maximum and minimum with a constraint

Name / Date _____ _____

PART A · IN CLASS — 48 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · DESMOS

The amount of a medicine in a patient's body is modeled by $A(t) = 640\left(\frac{1}{2}\right)^{\frac{t}{8}}$ milligrams, where t is the number of hours after the dose. After how many hours does the model estimate exactly 40 milligrams remain?

2. HARD · DESMOS

The function $h(t) = -16t^2 + 96t + 4$ models the height, in feet, of a ball t seconds after it is launched. The graph of $y = h(t)$ in the ty-plane has a vertex at $(3, 148)$. Which of the following is the best interpretation of the vertex in this context?

(A) The ball was launched from a height of 148 feet.

(B) The ball reached its maximum height of 148 feet 3 seconds after it was launched.

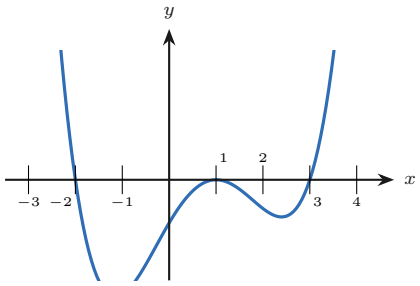
(C) The ball hit the ground 148 seconds after it was launched.

(D) The ball was 3 feet above the ground 148 seconds after it was launched.

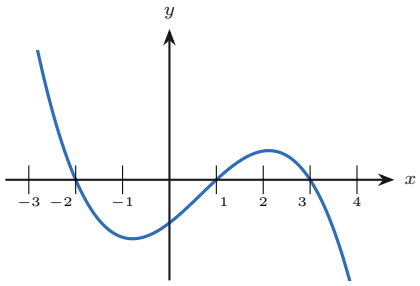
3. SAT Challenge · Desmos

Which of the following could be the graph of $y = -(x+2)(x-1)^2(x-3)$ in the xy-plane?

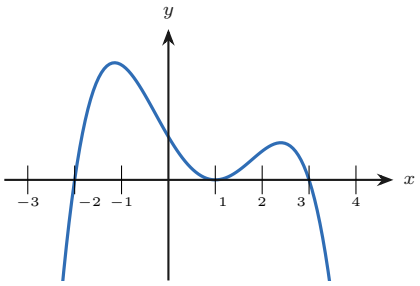
(A)



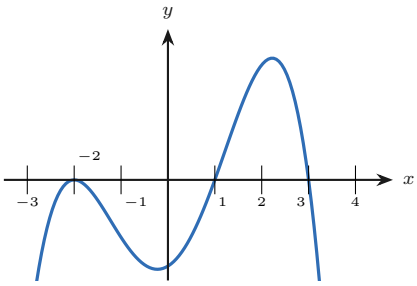
(B)



(C)



(D)



4. Hard · Desmos

A theater sells 600 tickets when the price is \$20. For each \$1 increase in price, 20 fewer tickets are sold. Which ticket price, in dollars, gives the greatest revenue?

(A) 25

(B) 12,000

(C) 12,500

(D) 5

5. HARD · DESMOS

The cost, in dollars, of making each unit when a factory produces n units is modeled by $C(n) = 0.02n^2 - 3.2n + 210$. What is the minimum cost per unit, in dollars, according to the model?

6. HARD · DESMOS

The quadratic function $f(x) = ax^2 + bx + c$, where a, b and c are constants, has two distinct real zeros. It is known that $f(2) = f(8)$, $f(0) < 0$ and $f(5) > 0$. Consider the following statements.

I. $b > 0$
II. The product of the two zeros is greater than 25.
III. Both zeros are positive.

Which of the statements must be true?

(A) I, II, and III

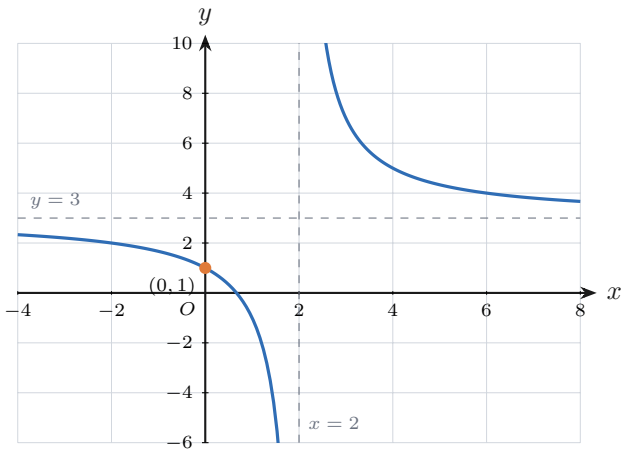
(B) I only

(C) II and III only

(D) I and III only

7. HARD · DESMOS

The graph of the rational function $f(x) = \dfrac{ax+b}{x-c}$ is shown, where a, b and c are constants. The dashed lines are the asymptotes $x = 2$ and $y = 3$, and the graph passes through the point $(0, 1)$. What is the value of $f(5)$? Enter your answer as a fraction.



8. HARD · DESMOS

The quadratic function f has its vertex at $(3, -2)$, and $f(5) = 6$. The function g is defined by $g(x) = \dfrac{3}{4} f\left(\dfrac{1}{2}x - 1\right)$. The graph of $y = f(x)$ has y-intercept $(0, p)$ and the graph of $y = g(x)$ has y-intercept $(0, q)$. What is the value of $q - p$?

(A) $\dfrac{45}{2}$

(B) $-\dfrac{23}{2}$

(C) $\dfrac{13}{2}$

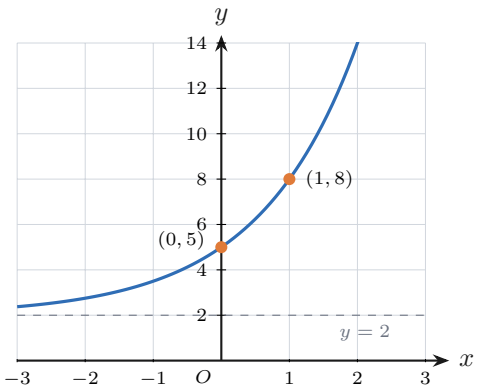
(D) -4

9. HARD · DESMOS

The function f is defined by $f(x) = 2^x + k$, where k is a constant. For a real number p, $f(p) = 9$ and $f(-p) = \frac{21}{4}$. What is the value of p?

10. HARD · DESMOS

The graph of the exponential function f is shown. The dashed line $y = 2$ is the horizontal asymptote, and the graph passes through the points $(0, 5)$ and $(1, 8)$. Which of the following could be an equation for f?



(A) $f(x) = 2^x + 4$

(B) $f(x) = 3 \cdot 2^x + 2$

(C) $f(x) = 2 \cdot 3^x + 3$

(D) $f(x) = 5 \cdot 2^x$

11. HARD · DESMOS

The altitude h, in meters, of a small rocket t seconds after launch is modeled by $h(t) = at^2 + bt + c$, where a, b, and c are constants. At launch the altitude is 3 meters; 2 seconds after launch it is 43 meters; and 5 seconds after launch it is 28 meters. According to the model, what is the altitude, in meters, 3 seconds after launch?

12. HARD · DESMOS

The quadratic function f has zeros at $x = -2$ and $x = 5$, and $f(1) = -24$. The function h is defined by $h(x) = f(x + 2) - 3f(-x)$. What is the value of $h(1)$?

13. HARD · DESMOS

The quadratic function f has an axis of symmetry at $x = 2$, and its graph passes through $(3, -3)$ and $(5, 13)$. The functions g and h are defined by $g(x) = \dfrac{x - 6}{2}$ and $h(x) = 3 - f(g(x))$. What is the y-coordinate of the y-intercept of the graph of $y = h(x)$?

14. SAT Challenge · Desmos

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. Some values of f are shown in the table below.

x	1	4
$f(x)$	54	16

What is the value of $f(-2)$?

15. Hard · Desmos

The function f is defined by $f(x) = c \cdot b^{x+3}$, where c and b are positive constants. If $f(1) = 1296$ and $f(3) = 729$, then for each increase of 1 in the value of x, $f(x)$ decreases by $p\%$. What is the value of p?

16. Hard · Desmos

The value V, in dollars, of a delivery van x years after it is purchased is modeled by $V(x) = 18{,}000(0.49)^{x/2}$. Which of the following best describes how the van's value changes each year?

(A) It decreases by 70% each year.

(B) It decreases by 51% each year.

(C) It decreases by 24.5% each year.

(D) It decreases by 30% each year.

17. HARD · NO CALCULATOR NEEDED

A sample of a radioactive isotope has a mass of 80 milligrams. The half-life of the isotope is 12 days. Which of the following functions gives the mass, in milligrams, of the sample w weeks after the sample is measured?

(A) $m(w) = 80\left(\frac{1}{2}\right)^{7w/12}$

(B) $m(w) = 80\left(\frac{1}{2}\right)^{12w/7}$

(C) $m(w) = 80\left(\frac{1}{2}\right)^{84w}$

(D) $m(w) = 80\left(\frac{1}{2}\right)^{w/12}$

18. HARD · DESMOS

In the xy-plane, the graph of the quadratic function p and the graph of the line $y = 5 - x$ intersect at exactly two points, one with x-coordinate 1 and one with x-coordinate 4. If $p(0) = -3$, what is the value of $p(6)$?

19. HARD · DESMOS

The function f is defined by $f(x) = ab^x + c$, where a, b, and c are constants and $b > 1$. The values $f(2) = 14$, $f(4) = 20$, and $f(6) = 32$ are known. What is the value of c?

20. HARD · DESMOS

For $x \geq 0$, let $f(x) = 120 - 90(0.8)^x$ and $g(x) = -3(x-4)^2 + 125$. Which of the following statements is true?

(A) Only f attains a maximum, 120

(B) Only g attains a maximum, 125

(C) Neither attains a maximum

(D) Both f and g attain a maximum

21. HARD · DESMOS

The polynomial $p(x) = x^3 + ax^2 + bx - 6$ is divisible by $x - 2$, and the remainder when $p(x)$ is divided by $x + 1$ is 12. What is the value of $a + b$?

22. HARD · DESMOS

$$-2(x-5)^2 + k = m(x-1)(x-r)$$

In the equation above, k, m, and r are constants. The equation is true for all real values of x. What is the value of k?

(A) -32

(B) 18

(C) 32

(D) 50

23. Hard · Desmos

A patient takes a 100-milligram dose of a medication at 8:00 a.m. and again at 8:00 p.m. every day, beginning at 8:00 a.m. on Monday. The dose is absorbed instantly, and between any two consecutive doses exactly half of the medication present in the body is eliminated. To the nearest tenth of a milligram, how much medication is in the body immediately after the 8:00 a.m. dose on Thursday?

(A) 198.4

(B) 199.2

(C) 187.5

(D) 196.9

24. SAT Challenge · Desmos

In the xy-plane, the line ℓ touches the parabola $y = x^2 - 4x + 7$ at exactly one point, the point where $x = 3$. What is the y-intercept of line ℓ?

25. Hard · Desmos

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. For every value of x, $f(x + 2)$ is 44% greater than $f(x)$. For every value of x, $f(x + 3)$ is p% greater than $f(x)$. What is the value of p?

(A) 132

(B) 60

(C) 66

(D) 72.8

26. HARD · DESMOS

The quadratic function f has a zero at $x = 2$. The graph of $y = f(x)$ in the xy-plane passes through the points $(0, -16)$ and $(5, 54)$. What is the minimum value of f?

(A) -9

(B) -1

(C) -18

(D) -16

27. HARD · DESMOS

The graph of the quadratic function f in the xy-plane passes through the points $(0, 1)$, $(1, -2)$ and $(3, 4)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ 3 units to the right and 5 units down. What is the y-coordinate of the y-intercept of the graph of g?

(A) -1

(B) 29

(C) 34

(D) 39

28. SAT CHALLENGE · DESMOS

$$\sqrt{x+7} - \sqrt{x-5} = 2$$

What is the solution to the equation above?

29. HARD · DESMOS

The function f is linear. The function g is defined by $g(x) = \dfrac{f(x+2) - x}{x - 3}$ for $x \neq 3$. Some values of g are shown in the table below.

x	1	5
$g(x)$	3	-11

What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

(A) -3

(B) -5

(C) -4

(D) 4

30. HARD · DESMOS

The function f is defined for $x \geq 0$ by

$$f(x) = 48\left(\frac{1}{8}\right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x \geq 0$ and displays the value of $f(0)$ as a constant factor?

(A) $f(x) = 12 \cdot 2^x$

(B) $f(x) = 48 \cdot 2^x$

(C) $f(x) = 12 \cdot \left(\dfrac{1}{2}\right)^x$

(D) $f(x) = 192 \cdot \left(\dfrac{1}{2}\right)^x$

31. Hard · Desmos

$$\frac{x}{x-2}+\frac{3}{x+2}=\frac{8}{x^2-4}$$

What is the sum of all real solutions of the equation above?

32. Hard · Desmos

The function $g(x)=2x^2+bx+c$, where b and c are constants, has zeros $k+1$ and $3k-5$, where k is a constant. If $g(2)=g(10)$, what is the value of c?

(A) 35

(B) 70

(C) 140

(D) 24

33. Hard · Desmos

Let $g(x)=2^x$ and $h(x)=3x-1$. For which values of x is $g(x)>h(x)$?

(A) $x>3$ only

(B) $x<3$

(C) $x<1$ or $x>3$

(D) $1<x<3$

34. HARD · DESMOS

Let $g(x) = 7 + 9\left(\frac{1}{4}\right)^x$ for all real numbers x. Consider the following statements.

I. $g(x) > 7$ for every real number x.
II. The minimum value of g is 7.
III. g is a decreasing function.

Which of the statements must be true?

(A) I and III only

(B) I and II only

(C) II and III only

(D) I, II, and III

35. SAT CHALLENGE · DESMOS

The function f is defined by $f(x) = \dfrac{2x+1}{x-3}$ for all $x \neq 3$, and f^{-1} is the inverse function of f. What is the value of $f^{-1}\big(f^{-1}(5)\big)$?

36. HARD · DESMOS

The temperature T, in degrees Fahrenheit, of a sample t minutes after it is placed in a chamber is modeled by one of the functions below for $t \geq 0$. For which model is 85 the greatest temperature the sample actually reaches?

(A) $T = 85 - 60(0.8)^t$

(B) $T = 40 + 45(1.05)^t$

(C) $T = 25 + 85(0.8)^t$

(D) $T = 25 + 60(0.8)^t$

37. HARD · DESMOS

The quadratic function $h(x) = ax^2 + bx + c$, where a, b and c are constants, satisfies $h(-3) = h(1)$, $h(0) > h(-1)$ and $h(-1) > 0$. Consider the following statements.

I. $b > 0$
II. $c > 0$
III. The equation $h(x) = 0$ has two distinct real solutions.

Which of the statements must be true?

(A) I, II, and III

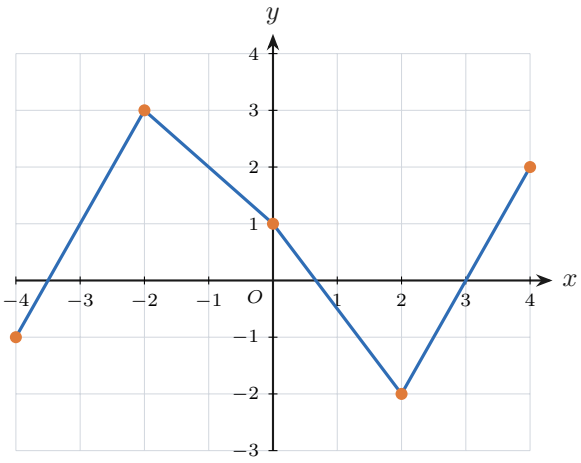
(B) I only

(C) I and II only

(D) II and III only

38. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown for $-4 \leq x \leq 4$; the function f is not defined for other values of x. The function g is defined by $g(x) = f(x - 1) - 1$. For how many values of x is $g(x) = 0$?



(A) 2

(B) 3

(C) 4

(D) 1

39. Hard · Desmos

The quadratic function f has its vertex at $(3, 4)$. The function g is defined by $g(x) = -2f(x+1)+7$, and the graph of $y = g(x)$ passes through the point $(4, 2)$. What is the value of $f(9) - f(2)$?

40. Hard · Desmos

A rocket's height, in feet, t seconds after launch is modeled by $h(t) = -16t^2 + bt + c$, where b and c are constants. The rocket is launched from a height of at most 94 feet, and the maximum height it reaches is 130 feet. For how many seconds is the rocket at a height of at least 94 feet?

41. Hard · Desmos

An investment grows exponentially, increasing by 6% every half-year. If V_0 is its initial value and $V(t)$ is its value after t years, after how many years will the value first reach three times the initial value? Round your answer to the nearest hundredth.

42. Hard · Desmos

A bakery's daily profit, in dollars, from baking n batches of bread is modeled by $P(n) = -2n^2 + 120n - 500$. The ovens can bake at most 25 batches per day. According to the model, what is the greatest daily profit the bakery can earn?

43. Hard · Desmos

The quadratic function $f(x) = 2x^2 + bx + c$ has $(x - 9)$ as a factor. The function g is defined by $g(x) = f(x + 3)$, and the graph of g in the xy-plane has its vertex at $x = 1$. What is the other zero of f?

44. Hard · Desmos

The quadratic function f satisfies $f(-1) = f(7)$, $f(0) = -4$ and $f(4) = 12$. What is the value of $f(10)$?

(A) 0

(B) 98

(C) -98

(D) -84

45. SAT Challenge · Desmos

The function g is defined by $g(x) = \left|x^2 - 4x\right|$. Consider the following statements.

I. $g(x) \geq 0$ for every real number x.
II. The graph of $y = g(x)$ has exactly two x-intercepts.
III. g is increasing for every $x > 2$.

Which of the statements must be true?

(A) I and II only

(B) II and III only

(C) I, II, and III

(D) I only

46. HARD · DESMOS

The table gives values of a quadratic function g at equally spaced inputs.

x	2	5	8	11
$g(x)$	44	95	200	359

What is the value of $g(17)$?

(A) 785

(B) 839

(C) 677

(D) 572

47. HARD · DESMOS

The concentration of a medication in the bloodstream, in milligrams per liter, is modeled by $C(t) = 18\left(\frac{1}{3}\right)^{\frac{t-4}{2}}$, where $t \geq 0$ is the number of hours after the first measurement. Which of the following is an equivalent form of $C(t)$ in which the maximum value of C (for $t \geq 0$) appears explicitly as the coefficient of the exponential term?

(A) $486 \cdot 3^{-\frac{t+2}{2}}$

(B) $18 \cdot 3^{\frac{4-t}{2}}$

(C) $162 \cdot 3^{-\frac{t}{2}}$

(D) $54 \cdot 3^{\frac{2-t}{2}}$

48. HARD · DESMOS

A water balloon is launched upward from a rooftop. Its height above the ground, in feet, t seconds after launch is modeled by a quadratic function h. The maximum height of 125 feet is reached at a time between $t = 1$ and $t = 6$, and $h(1) = 80$ and $h(6) = 105$. What is the height, in feet, of the balloon at the moment of launch?

(A) 45

(B) 75

(C) 37

(D) 80

PART B · SELF-ASSESSMENT — 25 PROBLEMS

49. HARD · DESMOS

The function f is defined by $f(x) = x^2 - 4x + 1$. The function g is defined by $g(x) = f(x + 3)$. What are the coordinates of the vertex of the graph of g in the xy-plane?

(A) $(2, 0)$

(B) $(-1, 0)$

(C) $(5, -3)$

(D) $(-1, -3)$

50. HARD · DESMOS

Selected values of the function g are shown in the table below.

x	1	2	3	4	5
$g(x)$	5	8	13	20	29

Which of the following could define g?

(A) $g(x) = 2^x + 3$

(B) $g(x) = 3x + 2$

(C) $g(x) = x^2 + 4$

(D) $g(x) = 3 \cdot 2^{x-1} + 2$

51. HARD · DESMOS

The functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2)) - g(f(2))$?

52. HARD · DESMOS

The value of an investment triples every 5 years. The value grows by the same percent every quarter (every 3 months). Which of the following is closest to that quarterly percent increase?

(A) 10.0%

(B) 5.6%

(C) 6.1%

(D) 24.6%

53. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 6$. Which of the following could be the x-intercepts of the graph of $y = f(2x)$?

(A) $x = 0$ and $x = 8$

(B) $x = -2$ and $x = 6$

(C) $x = -4$ and $x = 12$

(D) $x = -1$ and $x = 3$

54. HARD · DESMOS

A culture starts with 200 cells. The number of cells triples every 40 minutes. After how many hours will the culture contain exactly 16,200 cells? Enter your answer as a fraction.

55. HARD · NO CALCULATOR NEEDED

The quadratic function f has a minimum value of -5 at $x = 4$. The function g is defined by $g(x) = -2f(x-3)+1$. What is the maximum value of g?

56. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 2|x-2|-3$. What is the sum of the x-intercepts of the graph of $y = f(x)$ in the xy-plane?

(A) 4

(B) $\frac{7}{2}$

(C) 5

(D) 3

57. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = |x-3|$ and the function g is defined by $g(x) = f(x+1)-2$. What are the coordinates of the vertex of the graph of g?

(A) $(4,-2)$

(B) $(2,-2)$

(C) $(2,2)$

(D) $(4,2)$

58. Hard · No calculator needed

$$\frac{6}{x-3}+2=\frac{x+3}{x-3}$$

Which of the following describes the real solutions of the equation above?

(A) There are exactly two real solutions

(B) There are infinitely many real solutions

(C) There are no real solutions

(D) There is exactly one real solution

59. Hard · Desmos

The function f is defined by $f(x)=2x+1$. The function g is defined by $g(x)=f(f(x))-f(x)$. What is the value of $g(3)$?

60. Hard · Desmos

The function f is defined by $f(x)=(x-1)^2+4$. The function g is defined by $g(x)=f(x-k)+m$, where k and m are constants. The vertex of the graph of g is $(6,-3)$. What is the value of $k+m$?

61. HARD · DESMOS

The function f is defined by $f(x) = x^2 - 2x$. For how many real values of x is $f(f(x)) = 0$?

(A) 4

(B) 1

(C) 2

(D) 3

62. HARD · DESMOS

The functions f and g are defined by $f(x) = x + 3$ and $g(x) = 2x - 1$. For which values of x is $f(g(x)) = g(f(x))$?

(A) Only $x = 0$

(B) Only $x = 3$

(C) All real numbers

(D) No real numbers

63. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane passes through the point $(6, -2)$. The function g is defined by $g(x) = 2f(3 - x) + 1$. Which of the following points must lie on the graph of $y = g(x)$?

(A) $(3, -3)$

(B) $(-3, -5)$

(C) $(-3, -3)$

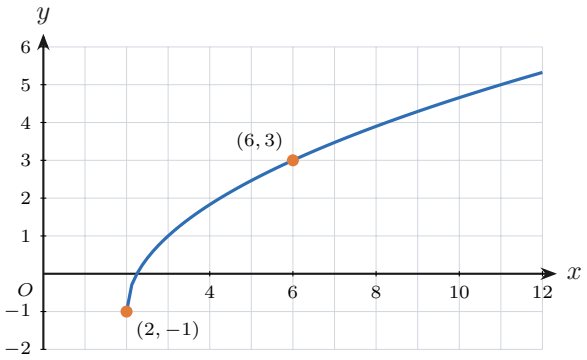
(D) $(9, -3)$

64. HARD · NO CALCULATOR NEEDED

The function f satisfies $f(1) = 3$ and $f(3) = 9$. The functions g and h are defined by $g(x) = f(x) + 4$ and $h(x) = f(x - 2)$. What is the value of $g(3) + h(3)$?

65. HARD · DESMOS

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown, where a, h and k are constants. The graph starts at the point $(2, -1)$ and passes through the point $(6, 3)$. What is the value of $f(11)$?



66. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane has its vertex at $(-2, 5)$. What are the coordinates of the vertex of the graph of $y = -f(x + 1) + 3$?

(A) $(-1, 8)$

(B) $(-3, -2)$

(C) $(-1, -2)$

(D) $(-3, 8)$

67. Hard · Desmos

A store sells $300 - 4p$ units of a product per week when the price is p dollars. The store only sets prices that are multiples of 5 dollars. What is the greatest weekly revenue, in dollars, the store can earn?

(A) 5,500

(B) 5,400

(C) 5,625

(D) 5,600

68. Hard · Desmos

The graph of $g(x) = (x - h)^2 + k$ in the xy-plane has a minimum value of 2 and passes through the points $(1, 6)$ and $(5, 6)$. What is the value of $h + k$?

69. Hard · No calculator needed

The function f is defined by $f(x) = x^4 - 13x^2 + 36$. Which of the following is an equivalent form of $f(x)$ in which the zeros of f appear as constants?

(A) $f(x) = (x - 2)(x + 2)(x - 3)(x + 3)$

(B) $f(x) = (x^2 - 4)(x^2 - 9)$

(C) $f(x) = (x - 1)(x + 1)(x - 6)(x + 6)$

(D) $f(x) = (x - 2)^2(x + 3)^2$

70. Hard · Desmos

The polynomial $p(x) = x^3 + bx^2 + cx + 12$ has integer coefficients b and c, and $x + 3$ is a factor of $p(x)$. If p has exactly one real zero, what is the greatest possible value of c?

71. Hard · No calculator needed

For all real numbers x, $4 \cdot 8^{(2x-1)/3} = a \cdot b^x$, where a and b are positive constants. What is the value of $a + b$?

72. Medium · Desmos

$$57x^2 + (57b + a)x + ab = 0$$

In the equation above, a and b are positive constants. The product of the solutions is kab, where k is a constant. What is the value of k? Enter your answer as a fraction.

73. HARD · DESMOS

$$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$$

In the equation above, a and b are constants. What is the value of ab?

(A) 20

(B) 24

(C) 40

(D) 18